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A Process For The Preparation ...

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A Process For The Preparation Of Photo Stable,Water Soluble Curcumin From Turmeric(curcuma Longa)

Abstract: The present invention relates to a process for the"p"reparalion of photo- stable, water-soluble curcumin from Turmeric spice (Curcuma longa) of Zingiberaceae family. Curcuminoids, the color principles of turmeric are soluble in organic solvents like acetone and ethyl acetate but insoluble in water at acidic or neutral pH. Curcuminoids are known to lose color upon exposure to air and light possibly due to oxidation reactions. The prior art shows preparation of curcumin derivatives of simple and complex sugar molecules and also of various metals to achieve higher solubility or dispersability of the pigment in aqueous based systems. However, in these cases the required stability of the pigments against oxidative and thermal stress is not achieved. In the present invention, curcumin has been rendered water-soluble by use of a suitable emulsifier and then selected additives were employed for the enhancement of photo-stability of the pigme

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encompasses the synergistic use of combination of TBHQ with piperine or ascorbyl palmitate for achieving the photo-stability of curcumin.

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Patent Information

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Applicants

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Claims

1. An improved process for the preparation of water-soluble pigment formulations possessing improved photo-stability from turmeric spice (Curcuma longa), wherein the steps comprise: (c) dissolving a water-soluble curcumin colorant in an emulsifier of the kind such as herein described followed by dilution with water to get a formulation of 0.33% of pigment content; (d) adding additives of the kind such as herein described to the formulation as obtained in step [a] at a ratio of 1: 1 ratio to obtain the desired photostable pigment formulation.

2. A process as claimed in claim 1, wherein the emulsifier is preferably selected from the group consisting of Polyoxyethylene sorbitan monostearate or any glyceryl ester of fatty acids.

3. A process as claimed in claim 1, wherein the additive used is preferably selected from the group consisting of piperine, TBHQ or ascorbyl palmitate either individually or in any combination.

4. An improved process for the preparation of water-soluble pigment formulations possessing improved photo-

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Specification

Field of invention

The present invention relates to a process for the preparation of water-soluble pigment formulations of improved photo-stability from Turmeric spice (Curcuma longa) of Zingiberaceae family.

Background of Invention and Description of Prior art

Turmeric is used in Indian cuisine mainly for its color and flavouring attributes. Colouring principles of turmeric are curcuminoids (curcumin, demethoxycurcumin and bis-demethoxycurcumin), of which curcumin is the major constituent. It is soluble in organic solvents like acetone and ethyl acetate and is practically insoluble in water at acidic or neutral pH. The pKa values for the dissociation of the three acidic protons in curcumin are 7.8, 8.5 and 9.0. Curcuminoids are known to lose color upon exposure to air and light possibly due to oxidation reactions. Degradation is dependent on the conditions of exposure and gets accelerated upon exposure to high heat and intense light conditions.

The prior art shows preparation of curcumin derivatives of simple and complex sugar molecules and also of various metals to achieve higher solubility or dispersibility of the pigment in aqueous based systems. However, in these cases the required stability of the pigments against oxidative and thermal stress is not achieved and often found to be disadvantageous from this point of view. Hence, there is need for development of a system to solubilize and preserve the color of turmeric when used in food formulations even when subjected to light and heat exposure.

The present invention relates to the preparation of a water-soluble curcumin colorant that is photo-stable.

The main uses of the invention are (i) the colorant would be soluble in water as well as in water-based formulations and foods (ii) and the colorant has higher stability when exposed to UV light.

Reference can be made to the US patent 4,368,208 (Schranz; John L, 1983) and GB2132205 (Schranz; John L, 1984) wherein a

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curcumin (to about 15%) and gelatin in an aqueous acetic acid solution for use in a wide variety of foodstuffs in either liquid or dry form. The drawback of the patent is the difficulty of removal of acetic acid, its cost, haziness at specific concentrations, and physical separation of curcumin from the dispersion.
Reference may be made to the U.S. Patent 4,307,117 (Leshik, R. R., 1981) wherein, a method of encapsulating a dispersion of curcumin so as to protect it from alkaline baking powders, and thereby provide stability to the color in mixes like dry cake powder, etc., is described. The drawback of the patent is that curcumin colorant is stabilized against color changes by maintaining the curcumin itself at mainly low, stable pH values from about 3.5 to about 4.5.
Reference can be made to the US patent 4,999,205 (Todd, Jr., Paul, H., 1991) wherein curcumin complexed with a water-soluble branched chain or cyclic polysaccharide or water-dispersible protein has been prepared by contacting the substrate and curcumin in an aqueous solution at a pH above 9 in red alkaline form, and then acidifying pH below 8, to obtain the curcumin complex in its neutral yellow form. The drawback of the patent is that the nature of the complex and its stability in various food systems are not well defined.

Reference can be made to the US patent 4,263,333 (Maing; Il Y. Miller; I, 1981), wherein preparation of water soluble Curcumin-metal color complexes capable of producing varying hues of the same colors for use in foodstuffs are described. The drawback of the patent is that such organometallic compounds are not permitted for use in food systems.
Reference can be made to the Patent, CN1543933 (Zhang qinglong, 2004) wherein a complex of curcumin and macromolecular complexing agent of hydroxyl propyl beta-cyclodextrin prepared and freeze dried for obtaining the curcumin injection, which can be used in the treatment for cancer, leukemia, hepatic diseases and cranial vascular diseases. The drawback of the patent is that the nature of the complex is not well defined for use in food systems.
Reference can be made to the patent CN1534013 (Wu xinhua, Zhang chuanlin, Zhang xiaoliang, 2004) wherein a process for preparing the water-soluble composite curcumin salt, which has the greatly increased biological utilization rate, is disclosed. The

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Reference can be made to the patent WO2004081023 (Jung In-Hak (KR); Kang Seung-Woo, Hong Sung-Yu, Kang Soo-Jung, Koh You-Seok, Chung Kwang-Hoe, 2004), wherein a method is described for preparation of a water-soluble curcumin by covalent-bonding of a sugar to the curcumin, thereby rendering the curcumin water-soluble for use in foods, soluble drugs and dyes. The drawback of the patent is that the nature of the compound and its toxicity status is not known for direct use in food systems

Reference can be made to the patent ES2121539 (Blasco Piquer Miguel, Martinez Fuentes Juan Gabriel, Faus Fortea Gerardo, Subitats Huerta Sebastian, 1998), wherein curcumin is dissolved in an alkaline aqueous solution of pH 11-14, previously heated to 30-40 degree C, to which water-soluble starch solution (10 and 20%) is added and allowed to react for 5 to 90 minutes at the temperature indicated and acidified to a pH between 8 and 5, the resulting paste being allowed to dry at 40 degree C for 48 to 72 hours to stabilize it in solid form, obtaining a yellow colouring product after grinding. The drawback of the patent is that the nature of the complex is not well defined for use in food systems.

Reference may be made to the U.S. patent 3,162,538 (Todd, 1964) wherein a microcrystalline suspension of curcumin and bixin (a red pigment from annatto) is prepared to color margarine and in which the curcumin prevented the bixin from turning pink on storage, when dissolved in the oil phase of the margarine. The drawback of the patent is that it does not aim at either rendering curcumin water-soluble or photostable.

Reference may be made the U.S. patent 3,455,838 (Marotta et al. 1969), wherein a modified starch is used for complexation of curcumin. The drawback of the patent is that in relation to curcumin content, these preparations will have low tinctorial power because the curcumin is crystalline, and the dispersions are hazy and precipitate.

Reference may be made to a U.S. patent, 4,138,212 (Stransky, C. E., 1979), wherein a process for making a curcumin coloring agent from turmeric root is reported by extracting the root with a hot soap solution at a pH of about 7 and filtering. This soap solution is dispersible in water, or it may be acidified to recover the curcumin mixed with fatty acids, and this used to color

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Reference may be made to the U.S. patent 4,263,333 (Maing et al. 1981), wherein the formation of water "soluble" metal adducts of curcumin dissolved in polysorbate 80, which provided hues different than curcumin itself when emulsified in water is described. The drawback of the patent is that while these different hues may be desired, they do not appear to be permitted for use in foods because they are organometallic compounds.

Reference may be made to the U.S. patent 4,414,234 (Sreenivasan, B and Baker, K. S., 1983), wherein the use of a redder natural pigment combined with a greenish yellow carotenoid as a margarine color, which would overcome the effect of curcumin in discoloring sulfite wrappers is described.

Woznicki et al. (U.S. Patent 4,475,919) in 1984, described a water-insoluble powder, such as cellulose (an unbranched polymer), with curcumin attached, made by adding an alkaline solution of curcumin to the cellulose suspended in aluminum chloride (an acidic solution), wherein the aluminum chloride precipitates the curcumin and assists in making it adhere to the cellulose. Optionally, organic solvents may be used to dissolve the curcumin prior to addition to the aluminum chloride solution. The disadvantage of this patent is that curcumin does not complex completely with the cellulose, and is readily removed with a solvent from curcumin.

Shinagawa et al., (JP, 79,163,866, 1979) described making a powder consisting of 1.6% curcumin and cellulose by mixing an aqueous suspension of the cellulose with an alkaline solution of curcumin, agitating, neutralizing, and filtering. The disadvantage of this is patent is that the curcumin sorbed, or precipitated on cellulose is neither soluble in water, nor does curcumin form stable complex with it.

Reference may be made to an Indian patent (Jaganmohan Rao et al., 401/Del/01), wherein deodorization of mother liquor of turmeric oleoresin is done to make a curcumin enriched base which is employed in the preparation of water-dispersible/soluble yellow colorant. The drawback of this patent is that the resultant formulation is not stable against exposure to light.

Reference may be made to the work of Tonnesen HH, Masson

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photochemical stability, Int J Pharm. 2002 Sep 5;244(1-2): 127-35), wherein cyclodextrin complexes of curcumin were prepared which resulted in an increase in water solubility at pH 5 and the hydrolytic stability of curcumin under alkaline conditions was strongly improved by complex formation. The drawback of this procedure is that the photodecomposition rate was increased compared to a curcumin solution in organic solvents as the cavity size and the charge and bulkiness of the cyclodextrin side-chains influenced the stability constant for complexation and the degradation rate of the curcumin molecule.

Reference may also be made to the work of Han G, Xu J, Li W, Ning C. (Study on preparation of the inclusion compound of curcumin with beta-cyclodextrin, Zhong Yao Cai, 2004 Dec;27(12):946-8), wherein inclusion compound of curcumin was prepared using the molar ratio between curcumin and beta-CD is 1:1at 40 degrees C for 2 h, to achieve increased (ten fold) solubility in water. The drawback of this procedure is that the photo-stability is not achieved.

Objects of the Invention

The main objective of the present invention is to provide a photo-stable and water- soluble formulation of curcuminoids from turmeric spice (Curcuma longa)', which obviates the

aforesaid drawbacks of the processes described in the prior art. Especially the prior art does
not deal with specific methods for stabilizing the pigments against degradation when
exposed to UV light.

Another object of the present invention is to provide an improved process for the preparation
of water-soluble and photostable pigments from Curcuma longa.

Yet another objective of the present invention is to provide a synergistic formulation of
additives comprising of piperine, T-Butylhydroquinone [TBHQ] and ascorbyl palmitate to
enhance the photostability of the prepared curcumins.

Summary of the Invention

The present invention provides a process for the preparation of water-soluble pigment formulations of improved photo-stability from turmeric (Curcuma longa), which comprises the preparation of a water-soluble curcumin colorant by dissolving it in an

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1 ratio in the resultant water-soluble preparation to achieve the desired photostable curcumins.
Accordingly, the present invention provides an improved process for the preparation of water-soluble pigment formulations possessing improved photo-stability from turmeric spice (Curcuma longa), wherein the steps comprise:
(a) dissolving a water-soluble curcumin colorant in an emulsifier of the kind such
as herein described followed by dilution with water to get a formulation of
0.33% of pigment content;
(b) adding additives of the kind such as herein described to the formulation as obtained in step [a] at a ratio of 1: 1 ratio to obtain the desired photostable pigment formulation.

Detailed Description of the Invention
The process employs solubilization of curcumin in water by a known process using a suitable emulsifier and then achieving the stability of the formulation against exposure to UV radiation by selection and inclusion of suitable antioxidants from piperine, ascorbyl palmitate and TBHQ. TBHQ is a well known antioxidant and its use in stabilization of vegetable oils is well documented in literature. However, Its efficacy as a color/photostabilizer in the stabilization of curcuminoids and especially in the presence of selected food additives has been demonstrated in the present patent.
In an embodiment of the invention, the additive used is piperine, T-Butylhydroquinone (TBHQ) or ascorbyl palmitate either individually or in any combination..
Novelty
The novelty lies in the use of selected additives for enhancement of the photo-stability of the pigment in the aqueous based systems, wherein the pigment is highly prone to photo-degradation.
The innovative step encompasses the invention of the synergistic combination of TBHQ with either piperine or ascorbyl palmitate for achieving the photo-stability of curcumin.
The following examples are given by way of illustration of the present invention and therefore should not be construed to limit the scope of the present invention.
Example 1

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Curcumin (92 % pure) isolated from turmeric oleoresin was used as the starting material for the preparation of water-soluble curcumin colorant. 0.1 g of curcumin was dissolved in Tween-60 (10 ml) and made up to 30 ml with water to get a formulation of 0.33 % of pigment content. The resultant water-soluble preparation (Formulation 1) was stored in glass test tube and its stability was studied under simulated accelerated conditions by exposing the colorant to UV light of 366 nm wavelength. The pigment stability in the sample was monitored at different time periods by suitable dilution with acetone, measurement of the absorbance at 425 nm and calculation of the retention of color.

Table 1 Curcumin color stability under exposure to UV light

(Table Removed)

Example 2

Curcumin (92 % pure) was used as the starting material for the preparation of water-soluble curcumin colorant with increased resistance to fading when exposed to UV light. 0.1 g of curcumin and 0.1 g of ascorbyl palmitate were dissolved in Tween-60 (10 ml) and made up to 30 ml with water to get a formulation of 0.33 % of pigment content and with ascorbyl

palmitate at 1: 1 ratio on pigment content. The resultant water-soluble preparation (Formulation 2) was stored in glass test tube and its stability was studied under simulated accelerated conditions by exposing the colorant to UV light of 366 nm wavelength. The pigment stability in the sample was monitored at different time periods by suitable dilution with acetone, measurement of the absorbance at 425 nm and calculation of the retention of color.

Table 2 Curcumin color stability in presence of Ascorbyl palmitate, under exposure to

UV light

(Table Removed)

Example 3

Curcumin (92 % pure) was used as the starting material for the preparation of water-soluble curcumin colorant with increased resistance to fading when exposed to UV light. 0.1 g of curcumin and 0.1 g of piperine were dissolved in Tween-60 (10 ml) and made up to 30 ml with water to get a formulation of 0.33 % of pigment content and with piperine at 1: 1 ratio on pigment content. The resultant water-soluble preparation (Formulation 3)

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glass test tube and its stability was studied under simulated accelerated conditions by exposing the colorant to UV light of 366 nm wavelength. The pigment stability in the sample was monitored at different time periods by suitable dilution with acetone, measurement of the absorbance at 425 nm and calculation of the retention of color.

Table3 Curcumin color stability in presence of piperine under exposure to UV light

(Table Removed)

Example 4

Curcumin (92 % pure) was used as the starting material for the preparation of water-soluble curcumin colorant with increased resistance to fading when exposed to UV light. 0.1 g of curcumin and 0.1 g of TBHQ were dissolved in Tween-60 (10 ml) and made up to 30 ml with water to get a formulation of 0.33 % of pigment content and with TBHQ at 1: 1 ratio on pigment content. The resultant water-soluble preparation (Formulation 4) was stored in glass test tube and its stability was studied under simulated accelerated conditions by exposing the colorant to UV light of 366 nm wavelength. The pigment stability in the sample was monitored at different time periods by suitable dilution with acetone, measurement of the absorbance at 425 nm and calculation of the retention of color.

Table 4 Curcumin color stability in presence of TBHQ under exposure to UV light

(Table Removed)

Curcumin (92 % pure) was used as the starting material for the preparation of water-soluble curcumin colorant with increased resistance to fading when exposed to UV light. 0.1 g of curcumin, 0.05 g of ascorbyl palmitate and 0.05 g of piperine were dissolved in Tween-60 (10 ml) and made up to 30 ml with water to get a formulation of 0.33 % of pigment content and with combination of additives at 1: 1 ratio on pigment content. The resultant water-soluble preparation (Formulation 5) was stored in glass test tube and its stability was studied under simulated accelerated conditions by exposing the colorant to UV light of 366 nm wavelength. The pigment stability in the sample was monitored at different time periods by suitable dilution with acetone, measurement of the absorbance at 425 nm and calculation of the retention of color.

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exposure to UV light
(Table Removed)
Example 6
Curcumin (92 % pure) was used as the starting material for the preparation of water-soluble curcumin colorant with increased resistance to fading when exposed to UV light. 0.1 g of curcumin, 0.05 g of ascorbyl palmitate and 0.05 g of TBHQ were dissolved in Tween-60 (10 ml) and made up to 30 ml with water to get a formulation of 0.33 % of pigment content and with combination of additives at 1: 1 ratio on pigment content. The resultant water-soluble preparation (Formulation 6) was stored in glass test tube and its stability was studied under simulated accelerated conditions by exposing the colorant to UV light of 366 nm wavelength. The pigment stability in the sample was monitored at different time periods by suitable dilution with acetone, measurement of the absorbance at 425 nm and calculation of the retention of color.
Table 6 Curcumin color stability in presence of Ascorbyl palmitate and TBHQ under
exposure to UV light
(Table Removed)
Example 7
Curcumin (92 % pure) was used as the starting material for the preparation of water-soluble curcumin colorant with increased resistance to fading when exposed to UV light. 0.1 g of curcumin, 0.05 g of piperine and 0.05 g of TBHQ were dissolved in Tween-60 (10 ml) and made up to 30 ml with water to get a formulation of 0.33 % of pigment content and with combination of additives at 1: 1 ratio on pigment content. The resultant water-soluble preparation (Formulation 7) was stored in glass test tubes and its stability was studied under simulated accelerated conditions by exposing the colorant to UV light of 366 nm wavelength. The pigment stability in the sample was monitored at different time periods by suitable dilution with acetone, measurement of the absorbance at 425 nm and calculation of the retention of color.
Table 7 Curcumin color stability in presence of TBHQ and piperine under exposure to UV light
(Table Removed)
Example 8
Curcumin (92 % pure) was used as the starting material for the preparation of water-soluble curcumin colorant with increased

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TBHQ were

dissolved in Tween-60 (10 ml) and made up to 30 ml with water to get a formulation of 0.33
% of pigment content and with combination of additives at 1: 1 ratio on pigment content. The
resultant water-soluble preparation (Formulation 8) was stored in glass test tubes and its
stability was studied under simulated accelerated conditions by exposing the colorant to UV
light of 366 nm wavelength. The pigment stability in the sample was monitored at different
time periods by suitable dilution with acetone, measurement of the absorbance at 425 nm
and calculation of the retention of color.
Table 8.Curcumin color stability in presence of Ascorbyl palmitate, TBHQ and piperine
under exposure to UV light
(Table Removed)
Advantages of the process are:
1. Makes use of an abundantly available edible raw material, namely curcuminoids for
colorant preparation
2. Overcomes the limitation of the turmeric pigments of not being soluble in water
3. Enhances the photo-stability of the formulation by a novel approach involving the
addition of TBHQ, ascorbyl palmitate and piperine
4. Increase in the pigment stability of the formulations when stored in diffused day light was noticed in comparison to control. However, the present proposal does not make any claims for enhancing the thermal stability of the curcumin pigments present in the formulations.

We claim:
1. An improved process for the preparation of water-soluble pigment formulations
possessing improved photo-stability from turmeric spice (Curcuma longa), wherein
the steps comprise:
(c) dissolving a water-soluble curcumin colorant in an emulsifier of the kind such as herein described followed by dilution with

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formulation as obtained in step [a] at a ratio of 1: 1 ratio to obtain the desired photostable pigment formulation.

2. A process as claimed in claim 1, wherein the emulsifier is preferably selected from the group consisting of Polyoxyethylene sorbitan monostearate or any glyceryl ester of fatty acids.
3. A process as claimed in claim 1, wherein the additive used is preferably selected from the group consisting of piperine, TBHQ or ascorbyl palmitate either individually or in any combination.
4. An improved process for the preparation of water-soluble pigment formulations possessing improved photo-stability from turmeric spice substantially as herein described with reference to the foregoing examples.

Documents

ORDERS

Section	Controller	Decision Date
15 in order case	Anita Jatav	2017-01-30
15	ANITA JATAV	2019-11-06
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